

NETWORK TRAFFIC ANALYSIS USING WIRESHARK

Cyber Security Internship Project Report

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Project area	Network Security and Packet Analysis
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This report documents the objective, methodology, protocol filters, security interpretation, and learning outcomes of a controlled Wireshark network-traffic analysis exercise.

1. Introduction

Network traffic analysis involves capturing and examining packets exchanged across a network. Wireshark provides visibility into protocol headers, endpoints, requests, responses, timing, and packet-level behaviour. In a controlled learning environment, this helps students understand normal communication and recognise indicators that may require further investigation.

2. Project Objectives

- Capture network traffic in an authorised environment.
- Apply display filters for DNS, HTTP, and ICMP traffic.
- Examine packet fields and protocol behaviour.
- Review protocol statistics and traffic distribution.
- Document security observations without exposing credentials or personal information.

3. Scope and Ethical Boundary

The exercise is limited to traffic generated by the student on a personally controlled or explicitly authorised device and network. The project does not involve intercepting another person’s communications, bypassing access controls, or attempting exploitation. Sensitive fields, authentication tokens, IP addresses, and personal information should be redacted from screenshots before publication.

4. Tools and Environment

Component	Purpose
Wireshark	Packet capture, filtering, and protocol inspection
Web browser	Generate authorised DNS and web traffic
Command Prompt / Terminal	Generate ICMP echo traffic with ping
GitHub	Publish report, screenshots, and project documentation

5. Methodology

5.1 Capture Preparation

- Open Wireshark and select the active authorised network interface.
- Start capture before generating test traffic.
- Avoid opening confidential applications during the capture.
- Generate only the traffic required for the exercise and stop capture promptly.

5.2 DNS Analysis

Open a known website and apply the display filter `dns`. Review query names, query types, transaction identifiers, response codes, and resolved addresses. Normal DNS activity generally contains a query followed by a matching response. Repeated failures, unexpected domains, or unusually frequent queries may warrant investigation.

5.3 HTTP Analysis

Where unencrypted HTTP test traffic is available in an authorised lab, apply `http`. Review request methods, host values, status codes, and content types. Most modern public websites use HTTPS, so an empty HTTP result is not necessarily an error. Encrypted TLS payloads should not be represented as readable HTTP content.

5.4 ICMP Analysis

Use a standard ping command against an authorised destination and apply `icmp` or `icmpv6`. Review echo request and echo reply pairs, identifiers, sequence numbers, packet size, and response timing. Missing replies can result from packet loss, routing issues, host unavailability, or firewall policy.

5.5 Protocol Statistics

Use Statistics → Protocol Hierarchy to review captured protocol distribution. Use Statistics → Conversations and Endpoints to understand communicating systems. These summaries help identify dominant protocols and unexpected communication patterns.

6. Display Filters Used

Filter	Use
<code>dns</code>	Display DNS queries and responses
<code>http</code>	Display unencrypted HTTP traffic
<code>tls</code>	Display TLS records and handshakes
<code>icmp</code>	Display IPv4 ICMP packets
<code>icmpv6</code>	Display IPv6 ICMP packets
<code>tcp.flags.syn == 1</code>	Review TCP connection-start packets

7. Analysis and Security Interpretation

7.1 DNS Traffic

DNS traffic demonstrates how a domain name is translated into an IP address. Important fields include query name, record type, response code, and returned records. Security review should focus on unexpected destinations, excessive failed queries, rapidly changing answers, or patterns inconsistent with normal browsing.

7.2 Web and TLS Traffic

HTTP exposes request metadata and content unless additional protection is used. HTTPS relies on TLS to protect application data in transit. Wireshark can still show connection metadata such as endpoints, ports, TLS versions, and handshake records, but encrypted application payloads are not normally readable.

7.3 ICMP Traffic

ICMP supports diagnostics and error reporting. Echo requests and replies demonstrate reachability. A small number of authorised ping packets is normal for this exercise. Sustained or unsolicited ICMP activity may require contextual review, but packet presence alone does not prove malicious behaviour.

7.4 Findings Summary

Area	Observation	Security meaning
DNS	Queries and responses can be correlated by transaction information.	Unexpected domains or repeated failures should be investigated.
HTTP/TLS	Modern browsing is commonly protected with TLS.	Encryption reduces exposure of application content in transit.
ICMP	Echo request and reply packets support reachability testing.	Unusual volume or pattern requires contextual investigation.
Statistics	Protocol Hierarchy summarises traffic composition.	Unexpected protocol dominance may guide deeper analysis.

Evidence note: Upload Mercy's own authorised Wireshark screenshots to the repository. Suggested evidence: DNS filter, ICMP filter, Protocol Hierarchy, and repository/project structure. Redact sensitive values before publishing.

8. Recommended Screenshot Evidence

File name	What the screenshot should show
01_dns_analysis.png	Wireshark with the dns display filter and selected query/response details.
02_icmp_analysis.png	Wireshark with icmp or icmpv6 packets produced by an authorised ping test.
03_protocol_hierarchy.png	Statistics → Protocol Hierarchy window showing traffic distribution.
04_repository_home.png	GitHub repository showing README.md, reports, screenshots, and LICENSE.

9. Learning Outcomes

- Understanding of packet capture and protocol layers.
- Practical use of DNS, HTTP/TLS, and ICMP display filters.
- Ability to inspect packet fields and protocol statistics.
- Awareness of privacy, authorisation, and responsible evidence handling.
- Experience documenting and publishing a cybersecurity project on GitHub.

10. Limitations

Results depend on the selected interface, network configuration, traffic generated, encryption, and capture privileges. HTTPS payloads remain encrypted without appropriate authorised keys. Wireshark observations provide indicators and context; a single packet or protocol alone should not be treated as proof of an attack.

11. Conclusion

This project builds foundational skills in network monitoring and packet analysis. By using controlled traffic and focused Wireshark filters, the exercise demonstrates how DNS, web, TLS, and ICMP communications can be examined and interpreted responsibly. The resulting report and screenshots provide clear evidence of cybersecurity learning for internship evaluation.

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